

xv6 Scheduling: MLFQ, FIFO and Round Robin

CS3.301 Operating Systems and Networks, Mini Project 1, Part 2

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Overview

This report covers the scheduler half of the mini project. I implemented a four level feedback queue (MLFQ) in xv6-riscv, added a non-preemptive FIFO scheduler so there was something to compare against, and kept the stock round robin (RR) scheduler as the default. A new system call, `waitx`, reports turnaround, waiting, response and running time for each child, and a user program called `schedulertest` generates a fixed workload with four different CPU burst lengths.

Section 1 walks through every change the specification asks about. Section 2 describes exactly what `schedulertest` does and which runs I made with it. Section 3 checks the MLFQ against a tick by tick trace. Section 4 compares the three schedulers on two workloads, and Section 5 lists every number that `schedulertest` printed in every run. All numbers in this report come from my own runs on a single CPU (`CPUS=1`), measured in timer ticks, and every plot was produced by `plot_mlfq.py`, which is submitted with the code.

1 Implementation Summary

1.1 Makefile and the SCHEDULER macro

I added a `SCHEDULER` variable to `xv6/Makefile`. Setting it to `MLFQ` or `FIFO` adds `-DMLFQ` or `-DFIFO` to `CFLAGS`, while leaving it unset (or setting it to `RR`) compiles the original round robin path with no scheduler macro at all. Any other value stops the build with an error, because a typo such as `SCHEDULER=mlfq` used to build round robin without complaint. I also added `TRACE=1`, which defines `MLFQTRACE` and makes an MLFQ kernel print one trace line per tick, and I added `schedulertest` to `UPROGS`. Since make does not notice when `CFLAGS` change, a `make clean` is needed between policies, otherwise old object files are reused.

```
make clean && make qemu                                # round robin (default)
make clean && make qemu SCHEDULER=MLFQ
make clean && make qemu SCHEDULER=FIFO
make clean && make qemu SCHEDULER=MLFQ TRACE=1 CPUS=1
```

1.2 Changes to struct proc

I added eight fields to `struct proc` in `kernel/proc.h`, all protected by `p->lock` just like `state`.

Field	Type	Meaning
<code>queue</code>	<code>int</code>	MLFQ priority, 0 is the highest and 3 the lowest
<code>slice_used</code>	<code>int</code>	ticks already used from the current queue's slice
<code>enter_seq</code>	<code>uint64</code>	ticket from a global counter; a smaller ticket is nearer the head of its queue
<code>ctime</code>	<code>int</code>	tick at which the process was created
<code>etime</code>	<code>int</code>	tick at which it exited (0 while alive)
<code>rtime</code>	<code>int</code>	ticks spent RUNNING
<code>wtime</code>	<code>int</code>	ticks spent RUNNABLE, which is time in the ready queue
<code>first_run</code>	<code>int</code>	tick of its first dispatch, or <code>-1</code> before that

There are no linked lists for the queues. A process's position is simply its queue number plus its ticket, and the scheduler treats "lowest queue, then lowest ticket" as the head. Pushing a process to the tail of a queue just means giving it a fresh ticket. I chose this because the fields live under the `p->lock` the kernel already takes while scanning the process table, so there is no extra queue lock to order against `p->lock` and `wait_lock`, which is exactly where schedulers like this tend to deadlock. The timing fields are compiled into every build because the comparison needs the same numbers from FIFO and RR. `NQUEUE` (4) and `BOOST_INTERVAL` (48) live in `kernel/param.h`.

1.3 Changes to `allocproc()`

`allocproc()` now places every new process in queue 0 with an empty slice, records `ctime = ticks`, zeroes `etime`, `rtime` and `wtime`, and sets `first_run` to `-1`. `freeproc()` clears the same fields so a recycled slot starts clean. The ticket itself is taken in `kfork()` and `userinit()` at the moment the process is marked RUNNABLE, through a small helper called `requeue()`. That way the child really joins the tail of queue 0 when it becomes ready, and not earlier, while `kfork()` is still copying memory. `kexit()` records `etime`. No explicit removal is needed on exit, since a ZOMBIE is never RUNNABLE and the scheduler never picks it.

1.4 Queue selection and preemption

Under MLFQ, `scheduler()` scans the whole process table and picks the RUNNABLE process in the lowest numbered queue, breaking ties with the smallest ticket. That gives strict priority between queues and FIFO order inside each one. Each lock is released during the scan, so after choosing, the scheduler locks the process again and checks that its state, queue and ticket have not changed; if another CPU or a boost got there first, it simply scans again.

For preemption, `usertrap()` and `kerneltrap()` in `kernel/trap.c` call `yield()` on a timer interrupt only when `mlfq_tick()` says so. That happens in two cases: the current process has used its whole slice, or `mlfq_higher_waiting()` finds a RUNNABLE process in a strictly higher queue. So a process running in queue 2 is preempted at the next tick boundary as soon as something arrives in queue 0, which is the example in the specification. A process preempted this way keeps its queue and its partly used slice, since it did nothing to deserve a demotion.

1.5 Time slice handling

Slices are {1, 4, 8, 16} ticks for queues 0 to 3. Once per tick, `clockintr()` on CPU 0 calls `update_time()`, which adds a tick to `rtime` and `slice_used` for every RUNNING process and to `wtime` for every RUNNABLE one. I placed this call before `ticks++` and `wakeup(&ticks)`, so a process that slept through the tick is counted as asleep rather than waiting. On the next timer interrupt of the CPU running the process, `mlfq_tick()` compares `slice_used` to the slice for its queue. When the slice is used up the process drops one queue (or stays in queue 3), gets a

fresh ticket at the tail and starts a new slice, which also makes queue 3 behave as round robin. On several CPUs the slice can run out between two of that CPU's own timer checks, so `sleep()` repeats the check before a process goes to sleep; without it, sleeping at that moment would dodge the demotion.

1.6 Voluntary yield handling

A process that sleeps on I/O, `pause()`, a pipe or `wait` leaves the ready set simply by not being `RUNNABLE`, so going to sleep needed no change. When `wakeup()` or `kkill()` makes it runnable again, it calls `requeue()`, which hands out a fresh ticket so the process joins the tail of the same queue it was in. Its priority is unchanged, as the rule requires. I also reset `slice_used` on wakeup. The specification does not say what happens to a partly used slice, and resetting it is the simpler reading, with one known cost: a process that always sleeps just before its slice ends keeps its priority. The trace in Section 3 shows exactly this for the burst 150 child.

1.7 Priority boosting

Every 48 ticks, `clockintr()` calls `mlfq_boost()`, which moves every live process (running, runnable, sleeping or still being set up) to queue 0 and clears its slice. `UNUSED` and `ZOMBIE` slots are skipped because they are not part of any queue. I left the tickets alone, so processes keep their relative order when they land in queue 0. The boost is what bounds starvation, both for CPU bound processes sitting in queue 3 and for any process kept down by a stream of new arrivals in queue 0.

1.8 procdump changes

Under MLFQ, Ctrl+P now prints each process's queue, its slice progress, its ticket, its running and waiting totals, and how many ticks remain until the next boost:

```
7 runble schedulertest  q2 slice 2/8 seq 212  run 34 wait 75  boost in 33
```

I used this constantly while developing to watch processes fall through the queues and jump back at a boost. The countdown to the next boost carries the same information as “ticks since the last boost” (48 minus the countdown). RR and FIFO builds keep the original output.

1.9 Supporting pieces

waitx. System call 23, `int waitx(int *turnaround, int *waiting, int *response, int *running)`, behaves like `wait` but also copies out four statistics, which are read before `freeproc()` clears them: `turnaround = etime - ctime`, `waiting = wtime`, `response = first_run - ctime` and `running = rtime`.

FIFO. Built with `SCHEDULER=FIFO`, the scheduler picks the `RUNNABLE` process with the smallest ticket, and the timer interrupt never calls `yield()`. Tickets are handed out whenever a process becomes runnable, so this is a proper ready queue: whoever became ready first runs next and keeps the CPU until it sleeps or exits.

Round robin. The policy is untouched: the timer yields every tick, so the quantum is one tick. The only additions on this path are the timing bookkeeping (`first_run` and `update_time()`), which never affects a scheduling decision.

2 The Test Program and the Runs

2.1 What `schedulertest` does

`schedulertest` is not part of stock xv6, so I wrote it (`xv6/user/schedulertest.c`) and added it to UPROGS. It is invoked as `schedulertest [nproc] [rounds]`, and it works in four steps.

1. **Arguments.** `nproc` is the number of children and `rounds` is a base round count. Missing, zero or negative values fall back to the defaults of 4 and 5. `nproc` is capped at 16 and `rounds` at 100.
2. **Forking the children.** Child i (counting from 0) is given the burst length `bursts[i % 4]` from the list {1, 40, 150, 600}, so the lengths repeat in that order. Its number of rounds is

$$my_rounds = \left\lfloor \frac{rounds \times (600 + 25)}{burst + 25} \right\rfloor,$$

where 25 is roughly the number of burst units in one tick under QEMU. A round is a burst plus about a tick of sleep, so this scaling gives the short bursters many more rounds and stops them from finishing long before the long ones.

3. **The work each child does.** A child runs `my_rounds` rounds of

```
burn(burst);    /* burst x 2,000,000 iterations of sink += i */
pause(1);      /* sleep for one tick, a voluntary yield */
```

and then calls `exit(0)`. `burn()` is pure CPU work with no system calls, so a child only leaves the CPU when its burst is over (by sleeping) or when the scheduler preempts it. The `pause(1)` is what makes every child give up the CPU voluntarily, as Section 2.3.2 of the specification asks.

4. **Collecting results.** The parent records each child's pid and burst, then calls `waitx` once per child. For every child that exits it prints

```
pid <pid> burst <b>: turnaround <T> waiting <W> response <R> running <run>
```

in the order the children finish, followed by one line with the averages of all four numbers, rounded to two decimal places.

The four burst lengths are chosen so that each one should settle in a different MLFQ queue, which is what makes the workload useful for all three schedulers: it mixes an interactive style job, two medium jobs and a CPU hog. Table 1 lists exactly what each child does in the two workloads I ran.

Child index	Burst units	Approx. ticks per burst	Intended queue	Rounds		Total burst units	
				4 5	8 3	4 5	8 3
0, 4	1	well under 1	Q0	120	72	120	72
1, 5	40	about 2	Q1	48	28	1920	1120
2, 6	150	about 7	Q2	17	10	2550	1500
3, 7	600	about 30	Q3	5	3	3000	1800

Table 1: The children created by `schedulertest`. With `schedulertest 4 5` only indices 0 to 3 exist; with `schedulertest 8 3` there are two children of each burst length. The tick estimates depend on the host, which is why I check the trace instead of relying on them.

2.2 Metrics

Every child’s numbers come from `waitx` and are counted in timer ticks.

- **Turnaround** $T = \text{etime} - \text{ctime}$: from the tick the child was created to the tick it exited.
- **Waiting** $W = \text{wtime}$: ticks the child spent `RUNNABLE`, that is, ready but not running.
- **Response** $R = \text{first_run} - \text{ctime}$: ticks from creation to its first dispatch.
- **Running** $= \text{rtime}$: ticks the child was `RUNNING`.

Because CPU 0 classifies every live process once per tick as running, runnable or asleep, the leftover $T - W - \text{running}$ is the child’s sleeping time. Each `pause(1)` should account for exactly one sleeping tick, which gives a strong consistency check: for every child, $T - W - \text{running}$ should equal its number of rounds. Section 5 shows that this holds exactly for all 28 children across the seven runs.

2.3 The runs

Table 2 lists every test run whose output appears in this report. Each kernel was built after a `make clean`, booted fresh, and run with one CPU so that only one process could run at a time.

Run	Build	Command	Children	Used for
1	SCHEDULER=MLFQ CPUS=1	<code>schedulertest 4 5</code>	4	comparison (Table 5)
2	SCHEDULER=MLFQ CPUS=1	<code>schedulertest 8 3</code>	8	comparison
3	CPUS=1 (round robin)	<code>schedulertest 4 5</code>	4	comparison
4	CPUS=1 (round robin)	<code>schedulertest 8 3</code>	8	comparison
5	SCHEDULER=FIFO CPUS=1	<code>schedulertest 4 5</code>	4	comparison
6	SCHEDULER=FIFO CPUS=1	<code>schedulertest 8 3</code>	8	comparison
7	SCHEDULER=MLFQ TRACE=1 CPUS=1	<code>schedulertest 4 5</code>	4	timeline only (Figure 1); tracing slows the kernel, so its times are not compared

Table 2: Every `schedulertest` run reported here.

Apart from `schedulertest`, I also ran xv6’s own test suite, `usertests -q`, under each scheduler to make sure the kernel changes did not break anything (Section 4.6).

3 MLFQ Analysis

3.1 How the timeline was produced

I built the kernel with `SCHEDULER=MLFQ TRACE=1 CPUS=1`, ran `schedulertest 4 5` and saved the console output. For every tick, CPU 0 prints one line per RUNNING or RUNNABLE process in the form `MLFQTRACE <tick> <pid> <queue> <running>`, and `plot_mlfq.py` turns those lines into Figure 1. Sleeping processes are not sampled, so they appear as gaps. The small vertical offset between pids is only there so overlapping lines stay visible. Tracing slows the kernel, so this run is used only for the timeline and never for timing numbers.

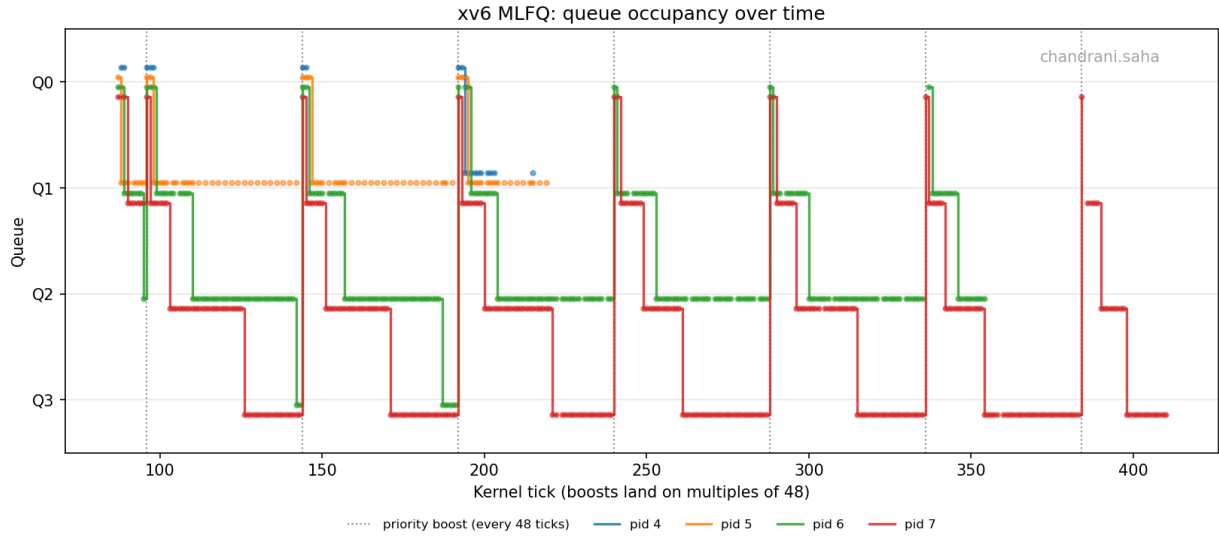


Figure 1: Queue of each `schedulertest` child over time (`schedulertest 4 5`, one CPU). pid 4 has a burst of 1 unit, pid 5 of 40, pid 6 of 150 and pid 7 of 600. Dotted lines mark the priority boost every 48 ticks.

3.2 What the trace shows

The workload ran from tick 87 to tick 410. I checked the trace line by line with a script, and the counts it gives match the kernel’s own `waitx` numbers for that run exactly (running 1, 51, 120 and 149 ticks, waiting 18, 35, 132 and 171 ticks for pids 4 to 7).

pid (burst)	Q0 run/wait	Q1 run/wait	Q2 run/wait	Q3 run/wait	Total run/wait	Sleeps seen
4 (1)	1 / 8	0 / 10	0 / 0	0 / 0	1 / 18	5
5 (40)	4 / 5	47 / 30	0 / 0	0 / 0	51 / 35	47
6 (150)	7 / 6	35 / 28	78 / 91	0 / 7	120 / 132	16
7 (600)	8 / 4	29 / 18	60 / 54	52 / 95	149 / 171	4

Table 3: Ticks spent running and waiting in each queue, counted from the trace. pid 4 is mostly invisible because most of its bursts start and end between two samples.

Transition	Slice (ticks)	Demotions seen	Running ticks before each one
Q0 to Q1	1	20	exactly 1 in all 20
Q1 to Q2	4	14	exactly 4 in all 14
Q2 to Q3	8	9	exactly 8 in all 9
Q3 to the tail of Q3	16	1	exactly 16

Table 4: Every demotion in the trace, counting slice use from the last wakeup (a wakeup starts a fresh slice).

A few more checks came out clean. On none of the 324 ticks was a runnable process waiting in a better queue than the one running, and no tick went by with the CPU idle while something was runnable. At each of the seven boosts from tick 96 to tick 384, every process present was in queue 0. Higher priority wakeups preempted a lower priority runner 44 times, for example pid 5 waking in queue 1 and displacing pid 7 in queue 2 at ticks 112 to 124.

3.3 Interpretation

pid 7 (burst 600) is the clearly CPU bound process. It slept only 4 times in 324 ticks, each of its descents from queue 0 used the full 1, 4 and 8 tick slices, and it spent 95 of its 171 waiting ticks in queue 3. pid 5 (burst 40) is I/O bound: it slept 47 times, almost always after about a tick of work, so it never left queue 1 (47 of its 51 running ticks) and kept jumping ahead of the longer jobs. pid 6 (burst 150) sits in between. It reached queue 3 twice, but after tick 221 it slept every 6 or 7 running ticks, and since a wakeup starts a fresh 8 tick slice it stayed in queue 2 and never actually ran in queue 3. The boost is what kept pid 7 from starving behind the others: it got 37 of its 149 running ticks (about 25%) in queues 0 and 1 right after boosts, and then sank back to queue 3 within 14 to 30 ticks each time.

4 Comparison Results

4.1 Method

I built each scheduler with `CPUS=1`, booted a fresh kernel and ran `schedulertest 4 5` and `schedulertest 8 3`, without tracing. Both workloads use the same four burst lengths; 8 3 simply has two children of each length and fewer rounds. Times are in ticks, and lower is better for every metric.

4.2 Averages

Scheduler	schedulertest 4 5			schedulertest 8 3		
	Turnaround	Waiting	Response	Turnaround	Waiting	Response
FIFO	265.50	148.00	2.00	308.63	236.25	17.75
RR	274.00	152.00	0.75	327.00	253.88	2.25
MLFQ	197.00	75.25	0.75	239.13	166.88	2.25

Table 5: Average turnaround, waiting and response time. The best value in each column is in bold.

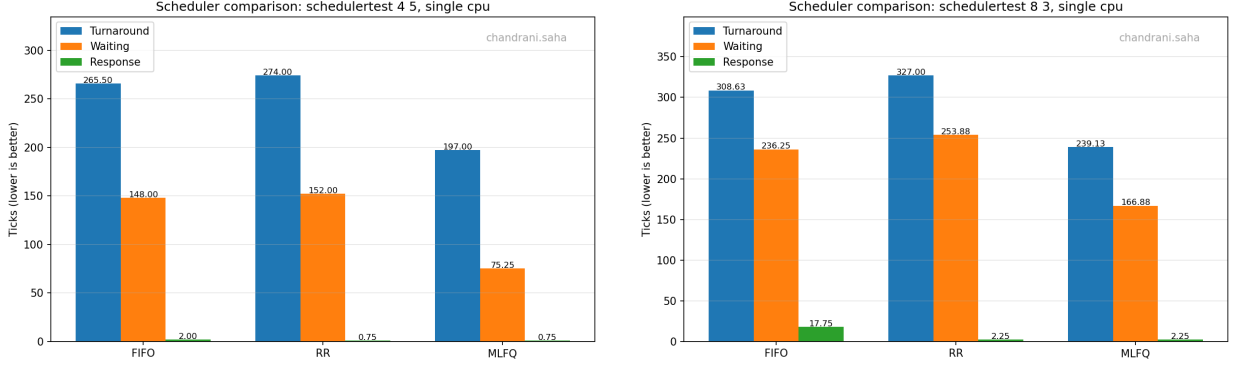


Figure 2: The averages from Table 5 as bar charts, for `schedulertest 4 5` (left) and `schedulertest 8 3` (right).

Comparison	4 5			8 3		
	Turnaround	Waiting	Response	Turnaround	Waiting	Response
MLFQ against RR	−28.1%	−50.5%	equal	−26.9%	−34.3%	equal
MLFQ against FIFO	−25.8%	−49.2%	−62.5%	−22.5%	−29.4%	−87.3%
FIFO against RR	−3.1%	−2.6%	2.7×	−5.6%	−6.9%	7.9×

Table 6: Relative differences between the averages. A negative percentage means the first scheduler is lower (better) by that much; $2.7\times$ means 2.7 times higher.

4.3 Who wins and who pays

Averages hide the most interesting part, which is how each policy treats short and long bursts. Table 7 and Figure 3 split the results by burst length.

Workload	Burst	MLFQ (T / W / R)	RR (T / W / R)	FIFO (T / W / R)
4 5	1	126 / 6 / 0	273 / 153 / 0	339 / 219 / 0
	40	126 / 29 / 0	233 / 135 / 0	311 / 202 / 0
	150	236 / 116 / 1	302 / 159 / 1	250 / 131 / 1
	600	300 / 150 / 2	288 / 161 / 2	162 / 40 / 7
8 3	1	82.0 / 10.0 / 1.5	328.0 / 256.0 / 1.5	389.0 / 314.5 / 15.5
	40	175.5 / 103.0 / 1.5	274.5 / 217.0 / 1.5	353.5 / 281.5 / 15.5
	150	353.5 / 284.5 / 2.5	361.0 / 275.0 / 2.5	294.0 / 225.5 / 17.0
	600	345.5 / 270.0 / 3.5	344.5 / 267.5 / 3.5	198.0 / 123.5 / 23.0

Table 7: Turnaround (T), waiting (W) and response (R) by burst length. For 8 3 each value is the mean of the two children with that burst.

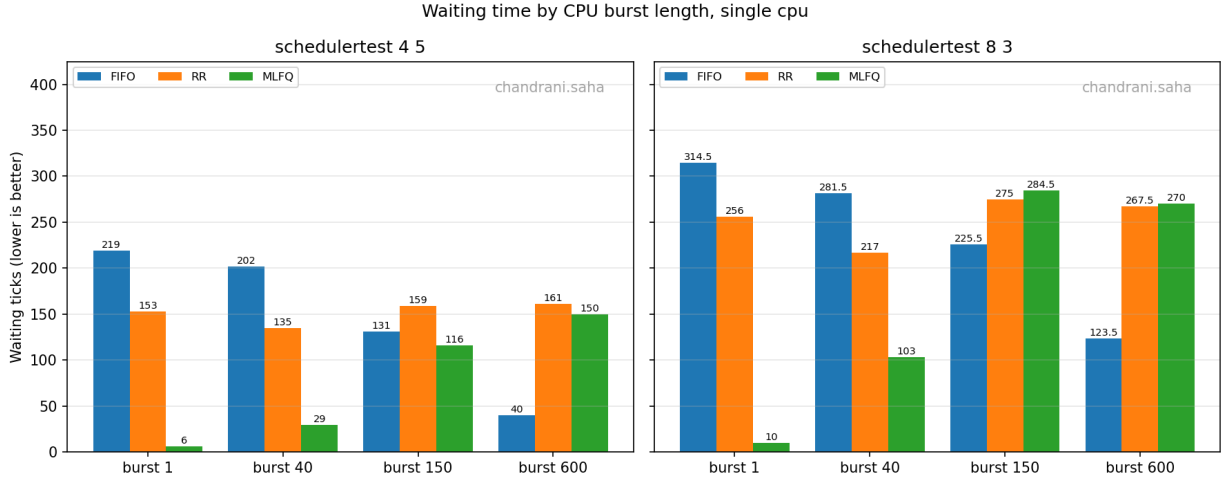


Figure 3: Waiting time for each burst length under each scheduler.

4.4 Discussion

MLFQ gave the lowest average turnaround and waiting time in both workloads, 197.00 and 75.25 ticks for 4 5 against 274.00 and 152.00 for RR. Almost all of that gain comes from the short bursters: a burst 1 child never uses up its one tick slice in queue 0, so it runs almost as soon as it wakes and waited just 6 ticks over 120 rounds, while under RR it waited 153 and under FIFO 219. RR had the worst waiting time in both workloads because its quantum is a single tick, so every wakeup has to queue behind a full tick of every other runnable job, and the burst 40 child, which needs a little over a tick, gets cut off and has to go round the whole queue again to finish. That is why RR’s waiting time depends so much on the quantum: a larger quantum would let those jobs finish in one go but make short jobs sit behind longer turns, and once the quantum is longer than every burst, RR becomes FIFO. FIFO shows the opposite trade-off, because nothing ever takes the CPU away: the burst 600 child finished first (turnaround 162 in 4 5) while the short jobs got stuck behind whole 24 to 30 tick bursts, the classic convoy effect. For response time, MLFQ and RR tied exactly, since each new child starts in queue 0 and waits at most one tick for every earlier child. FIFO looked fine in 4 5 (2.00) only because the long job happened to be forked last, but in 8 3 the second group of children had to wait for a whole 24 tick burst before their first dispatch, which pushed the average to 17.75, almost eight times RR’s. MLFQ’s gains are not free: the long jobs pay for them, and the burst 600 children waited 270 ticks in 8 3 compared with 123.5 under FIFO, mostly getting the CPU only after a boost.

4.5 Limitations

- Time is sampled once per tick on CPU 0, so a burst shorter than a tick is usually charged nothing. That is why the burst 1 child shows `running 0`, and a process sharing a tick can be charged the whole tick.
- Response time only counts the very first dispatch. After that, a burst 1 child under FIFO waits about 4 ticks per round in 8 3, which the response column never shows.
- Every configuration was run once, on a single CPU. The total CPU work agrees within 2% across schedulers for 8 3 (352 to 359 ticks), but FIFO’s 4 5 run recorded about 6% less than the other two (280 against 297 and 298), most likely because QEMU ran a little faster during that run. The gap between RR and FIFO (3 to 7%) is small enough that it should not be read too strongly; the MLFQ results are far outside that margin.

- Tracing is too slow to time, so the traced run (average turnaround 216.75) is used only for the timeline.

4.6 Correctness checks

Beyond the numbers, I ran xv6’s own `usertests -q`. It passes under MLFQ on one and on three CPUs, and under RR. FIFO hangs at the `preempt` test, as any non-preemptive scheduler must, because that test needs a spinning child to be taken off the CPU. Run on its own right after boot, `usertests reparent` can fail under MLFQ: `init` is demoted during boot and a steady stream of new queue 0 processes keeps it off the CPU until the first boost while zombies fill the process table. That is strict priority starving a low priority process, which the 48 tick boost bounds, and the test passes inside the full suite.

5 Complete schedulertest Output

This section lists every number `schedulertest` printed in the seven runs from Table 2, with nothing left out. Rows appear in the order the children finished, which is the order the program printed them. The **Rounds** column is taken from Table 1, and the **Sleep** column is $T - W - \text{running}$, which I computed. In every row of every run Sleep equals Rounds exactly, so each `pause(1)` was charged exactly one sleeping tick and the three counters account for the child’s whole lifetime with nothing missing and nothing counted twice. The last row of each table is the average line printed by the program.

pid	Burst	Rounds	Turnaround	Waiting	Response	Running	Sleep
4	1	120	126	6	0	0	120
5	40	48	126	29	0	49	48
6	150	17	236	116	1	103	17
7	600	5	300	150	2	145	5
Printed average			197.00	75.25	0.75	74.25	

Table 8: Run 1: MLFQ, `schedulertest 4 5`.

pid	Burst	Rounds	Turnaround	Waiting	Response	Running	Sleep
10	1	72	82	10	0	0	72
14	1	72	82	10	3	0	72
11	40	28	175	105	0	42	28
15	40	28	176	101	3	47	28
13	600	3	344	268	2	73	3
17	600	3	347	272	5	72	3
12	150	10	353	282	1	61	10
16	150	10	354	287	4	57	10
Printed average			239.13	166.88	2.25	44.00	

Table 9: Run 2: MLFQ, `schedulertest 8 3`.

pid	Burst	Rounds	Turnaround	Waiting	Response	Running	Sleep
5	40	48	233	135	0	50	48
4	1	120	273	153	0	0	120
7	600	5	288	161	2	122	5
6	150	17	302	159	1	126	17
Printed average			274.00	152.00	0.75	74.50	

Table 10: Run 3: round robin, `schedulertest 4 5`.

pid	Burst	Rounds	Turnaround	Waiting	Response	Running	Sleep
14	40	28	274	217	3	29	28
10	40	28	275	217	0	30	28
9	1	72	327	255	0	0	72
13	1	72	329	257	3	0	72
12	600	3	339	264	2	72	3
16	600	3	350	271	5	76	3
11	150	10	361	275	1	76	10
15	150	10	361	275	4	76	10
Printed average			327.00	253.88	2.25	44.88	

Table 11: Run 4: round robin, `schedulertest 8 3`.

pid	Burst	Rounds	Turnaround	Waiting	Response	Running	Sleep
7	600	5	162	40	7	117	5
6	150	17	250	131	1	102	17
5	40	48	311	202	0	61	48
4	1	120	339	219	0	0	120
Printed average			265.50	148.00	2.00	70.00	

Table 12: Run 5: FIFO, `schedulertest 4 5`.

pid	Burst	Rounds	Turnaround	Waiting	Response	Running	Sleep
12	600	3	194	119	7	72	3
16	600	3	202	128	39	71	3
11	150	10	293	226	1	57	10
15	150	10	295	225	33	60	10
10	40	28	353	283	0	42	28
14	40	28	354	280	31	46	28
9	1	72	381	306	0	3	72
13	1	72	397	323	31	2	72
Printed average			308.63	236.25	17.75	44.13	

Table 13: Run 6: FIFO, `schedulertest 8 3`. The average running figure was cut off when I copied the console output; 44.13 is computed from the eight rows above ($353 / 8 = 44.125$, which the program's rounding prints as 44.13).

pid	Burst	Rounds	Turnaround	Waiting	Response	Running	Sleep
5	40	48	134	35	0	51	48
4	1	120	139	18	0	1	120
6	150	17	269	132	1	120	17
7	600	5	325	171	2	149	5
Printed average			216.75	89.00	0.75	80.25	

Table 14: Run 7: MLFQ with tracing, `schedulertest 4 5`. Used only for the timeline.

5.1 Total CPU work per run

Since every scheduler runs the same children doing the same number of loop iterations, the summed running time should be about the same in each run with the same workload.

Workload	MLFQ	RR	FIFO
<code>schedulertest 4 5</code>	297	298	280
<code>schedulertest 8 3</code>	352	359	353

Table 15: Sum of running over all children, in ticks.

For 8 3 the three schedulers agree within 2%. For 4 5, FIFO recorded about 6% less than the other two. Per tick sampling can only shift the count by about one tick per burst, which is not enough to explain that gap, so I think QEMU simply ran a little faster during that run. This is the main reason I do not draw strong conclusions from the small gap between FIFO and round robin.

5.2 procdump snapshots from run 1

While run 1 was going, I pressed Ctrl+P twice. Both snapshots show the four children sitting in different queues, each with its slice progress staying within the slice for that queue.

```

4 sleep  schedulertest  q0 slice 0/1 seq 88  run 0  wait 2  boost in 26
5 run    schedulertest  q1 slice 0/4 seq 89  run 6  wait 9  boost in 26
6 runble schedulertest  q2 slice 0/8 seq 90  run 8  wait 11 boost in 26
7 runble schedulertest  q2 slice 1/8 seq 77  run 6  wait 14 boost in 26

4 run    schedulertest  q0 slice 0/1 seq 223 run 0  wait 6  boost in 33
5 runble schedulertest  q1 slice 1/4 seq 222 run 41 wait 29 boost in 33
6 runble schedulertest  q2 slice 0/8 seq 217 run 34 wait 72 boost in 33
7 runble schedulertest  q2 slice 2/8 seq 212 run 34 wait 75 boost in 33

```

In the second snapshot, pid 4 (burst 1) has already waited 6 ticks, which is its final waiting total in Table 8: from then until it exited it never had to wait again.

6 Conclusion

I added a four level feedback queue scheduler to xv6, along with a non-preemptive FIFO scheduler and timing bookkeeping that works under all three policies, so they could be compared fairly. The MLFQ follows every rule in the specification. New processes join the tail of queue 0, the highest non-empty queue always runs, and a process that uses its whole slice of 1, 4, 8 or 16 ticks drops a

level. A process that sleeps comes back to the tail of the same queue, and everything returns to queue 0 every 48 ticks. The queues need no linked lists or extra locks: a queue number and a ticket per process are enough, which kept the locking simple and safe on several CPUs.

The trace confirms that the implementation behaves exactly as designed. All 43 demotions happened after exactly 1, 4 or 8 running ticks, and the one re-queue inside queue 3 after exactly 16. In 324 ticks no runnable process ever waited behind a lower priority one, and every process present was back in queue 0 at all seven boosts. The trace also shows the classic MLFQ behaviour clearly: the burst 40 child kept its high priority by sleeping often, the burst 600 child sank to queue 3, and the boost was the only thing lifting it back up. The measurements are internally consistent. Across all 28 children in seven runs, turnaround minus waiting minus running equals the number of `pause(1)` calls exactly, and every printed average matches its rows.

The comparison shows three schedulers making three different trade-offs on the same workload.

- **MLFQ** was the best overall. It cut average waiting time by 50.5% against round robin and 49.2% against FIFO in `schedulertest 4 5`, and by 34.3% and 29.4% in `schedulertest 8 3`. It did this by favouring short bursts, which behaves much like shortest job first without knowing burst lengths in advance. The price is paid by the long jobs, which in `8 3` waited more than twice as long as under FIFO.
- **Round robin** was the fairest in the sense that every job got a turn within a few ticks, matching MLFQ's best response time (0.75 and 2.25 ticks). Its one tick quantum, however, made every job wait behind every other job on every burst, giving it the highest waiting time in both workloads.
- **FIFO** was the simplest and let the long job finish first, but short jobs got stuck behind whole 24 to 30 tick bursts. Its response time rose to 17.75 ticks in `8 3`, almost eight times that of the other two, and being non-preemptive it cannot pass the `usertests` preemption test.

The main caveats are that times are sampled once per tick, each configuration was run once on a single CPU, and the gap of 3 to 7% between FIFO and round robin is too small to rank them with confidence. The MLFQ advantage, on the other hand, is large and consistent across both workloads. Overall, the results show why MLFQ is a sensible default for a general purpose system. By watching how processes actually use the CPU, it gives interactive, short burst work the responsiveness of round robin and the throughput benefits of shortest job first, while the periodic boost keeps long running jobs from starving.